

CLOUD INFRASTRUCTURE · SEMINAR PRESENTATION

Amazon Lightsail

Platform-as-a-Service, Lite

How AWS packaged its own building blocks into a fixed-price virtual private server — and what you trade away to get that simplicity.

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30 Nov 2016

Launched at AWS re:Invent



\$5 / month

Entry bundle — compute, SSD and 1 TB of transfer in one price



8 resource types

Instances, containers, databases, storage, CDN, DNS and more

What this talk covers



The problem worth solving

Why AWS built a simpler front door



Anatomy of the service

Blueprints, bundles and the resource catalogue



Lightsail versus the field

Amazon EC2, DigitalOcean, Heroku, Azure



Limits and the exit

What you give up, and how you leave



Where Lightsail sits

IaaS, PaaS-Lite, PaaS and SaaS compared



The pricing model in full

Instances, databases, storage, networking



Hands-on and use cases

A production site in under ten minutes



Verdict

When this is the right tool

MOTIVATION

The problem: power arrived with complexity

Amazon EC2 gives you every knob a data centre has. For a team that only wants a website, a blog or a small application online, most of those knobs are a tax rather than a feature.



Too many decisions

Before a raw EC2 instance serves traffic you must design a VPC, a subnet, a route table, an internet gateway, a security group, an EBS volume, a key pair and an Elastic IP.



Unpredictable bills

Compute, storage, egress and the IP address are four separate meters. A small site cannot forecast next month's invoice.



Skills mismatch

Students, freelancers and small businesses need a running server, not a cloud architecture certification.

750+

Amazon EC2 instance types to choose between

8

Lightsail general-purpose bundles — the whole menu

1

Line on your bill, fixed before you deploy

Fewer choices, stated cost, identical AWS hardware.

Where Lightsail sits in the cloud stack

Why “PaaS-Lite”?

A true PaaS hides the operating system entirely — you push code and never meet a server.

Lightsail pre-builds the OS and the runtime for you, then hands you the root password anyway.

You get PaaS convenience on day one and IaaS control on day two. That hybrid position is the whole product.

	On-Premises your data centre	IaaS Amazon EC2	PaaS-Lite Amazon Lightsail	PaaS Beanstalk / Heroku	SaaS Microsoft 365
Data	You	You	You	You	AWS
Application	You	You	You	You	AWS
Runtime / Middleware	You	You	Shared	AWS	AWS
Operating System	You	You	Shared	AWS	AWS
Virtualisation	You	AWS	AWS	AWS	AWS
Servers / Compute	You	AWS	AWS	AWS	AWS
Storage	You	AWS	AWS	AWS	AWS
Networking	You	AWS	AWS	AWS	AWS

■ You manage
■ Shared — pre-built, still yours to change
■ AWS manages

Figure 1 — Division of management responsibility across cloud service models.

DEFINITION

What Amazon Lightsail actually is

Amazon Lightsail is AWS's virtual private server offering: pre-configured compute, storage, networking and managed services, sold as fixed-price monthly bundles through a deliberately simplified console.



Announced 30 November 2016

Launched at AWS re:Invent in US East (N. Virginia); expanded to Hong Kong, São Paulo and Spain in June 2026.



One price, four things

Every bundle folds vCPU, memory, SSD storage and a monthly data-transfer allowance into a single figure.



Blueprint, not blank box

Choose WordPress, LAMP, Node.js or a plain OS; the instance boots with the software already installed.



Real AWS underneath

Instances are EC2, disks are EBS, DNS is Route 53. Nothing is a proprietary dead end.

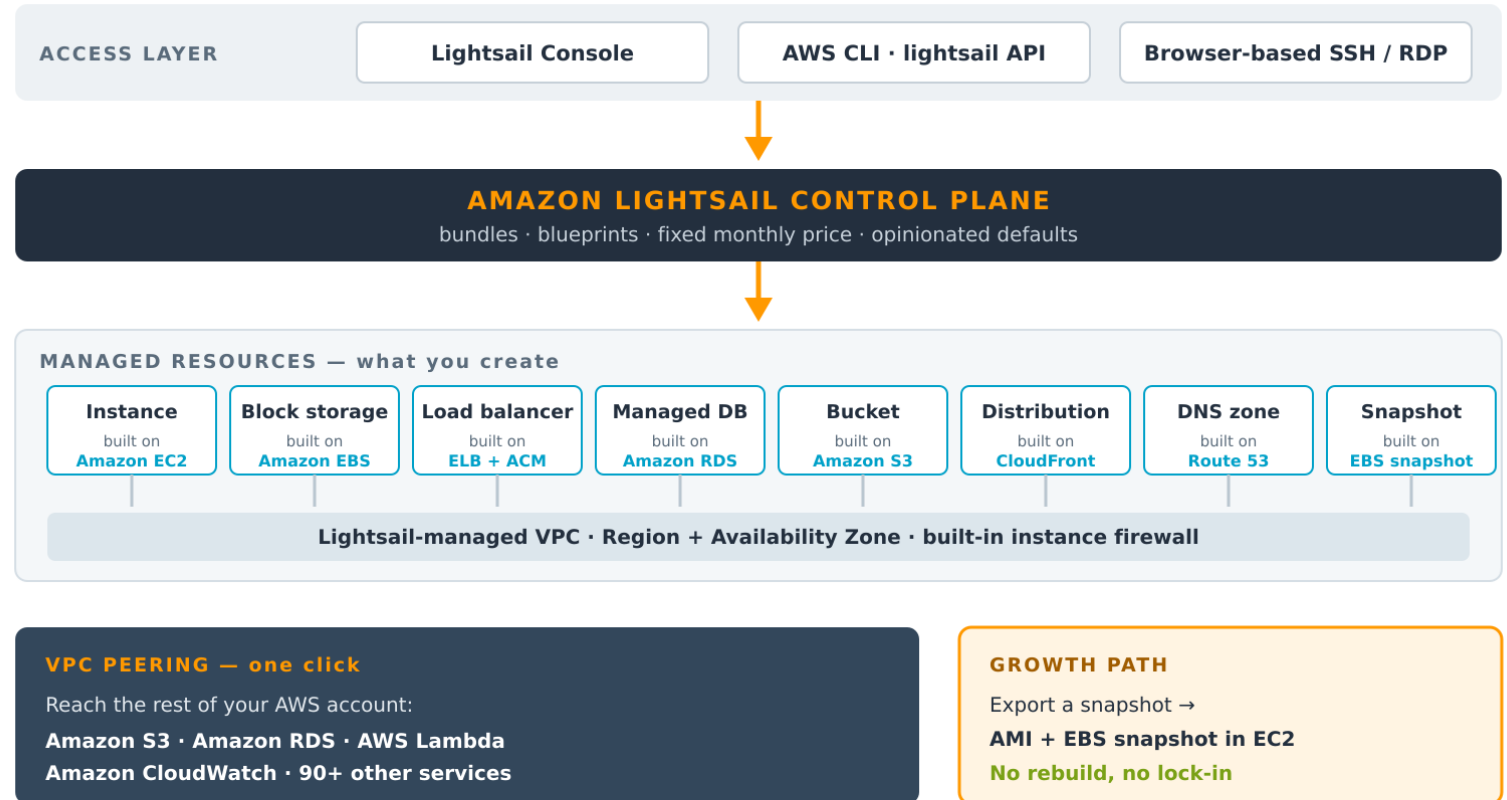
How the abstraction is built

The one idea

Lightsail is not a new infrastructure platform. It is a curated control plane over services AWS already sells.

- An instance is an EC2 instance
- A disk is an EBS volume
- A load balancer is ELB plus a free ACM certificate
- A distribution is CloudFront
- A DNS zone is Route 53

Simplicity is the feature; the hardware is unchanged.



Every Lightsail resource is a standard AWS primitive with the configuration surface hidden — **that hiding is the product. You trade knobs for a price you can predict.**

Figure 2 — The Lightsail control plane and the AWS primitives beneath it.

The resource catalogue

Ten resource types cover the whole surface of the service.

Resource	What it gives you	Built on
Instance	Virtual server with a fixed bundle of vCPU, RAM, SSD and transfer	Amazon EC2
Container service	Runs Docker images behind a managed HTTPS endpoint — no cluster to operate	ECS / Fargate
Managed database	MySQL or PostgreSQL with automated backups and an optional standby	Amazon RDS
Block storage disk	Additional SSD volumes attached to a running instance	Amazon EBS
Object storage bucket	Static assets, media and backups served over HTTPS	Amazon S3
Load balancer	Spreads traffic across instances; includes a free, auto-renewed certificate	ELB + ACM
CDN distribution	Edge caching in front of an instance, bucket or load balancer	Amazon CloudFront
DNS zone	Authoritative DNS with three million queries a month included	Amazon Route 53
Static IP	A fixed public IPv4 address that survives stop, start and restore	Elastic IP
Snapshot	Point-in-time backup of an instance or disk; exportable to EC2	EBS snapshot

Every row is an AWS service with its configuration surface deliberately reduced — which is exactly why a Lightsail workload can later be lifted onto the full service without a rewrite.

Two choices: the blueprint and the bundle



Blueprint — the software

What the machine boots into.

Operating systems

Amazon Linux 2 and 2023 · Ubuntu · Debian · AlmaLinux 9 · CentOS · FreeBSD · openSUSE · Windows Server 2016 / 2019 / 2022

Application stacks

WordPress · WordPress Multisite · LAMP · Nginx (LEMP) · Node.js · Ruby on Rails · Django · Drupal · Joomla · Magento · PrestaShop · Ghost · Redmine · GitLab · cPanel & WHM · Plesk



Bundle — the hardware

What it runs on, and what it costs.

- One bundle fixes vCPU, RAM, SSD size and the monthly transfer allowance together
- Three families: general purpose, memory optimised (to 512 GB RAM, Feb 2026) and compute optimised (to 72 vCPU, Apr 2026)
- Entry tiers use burstable CPU; \$24/month and above give dedicated vCPU
- Bundles can be resized upward by taking a snapshot and restoring onto a larger plan



Currently in transition — worth knowing for any new build

AWS stopped shipping newer Bitnami-packaged blueprints on 19 May 2026. WordPress, LAMP, Nginx and Node.js images packaged by Bitnami retire on 19 November 2026; Joomla, Magento, Drupal, Ghost, Django, GitLab, Redmine, MEAN and PrestaShop follow on 19 May 2027. AWS-maintained replacements for Node.js, LAMP and Ruby on Rails shipped in January 2026.

Instance bundles — the published price list

Linux/Unix general-purpose bundles with a public IPv4 address. Windows bundles carry a licence premium.

Bundle	Memory	vCPU	SSD storage	Transfer / month	Linux \$/month	Windows \$/month
Nano	512 MB	2	20 GB	1 TB	\$5	\$9.50
Micro	1 GB	2	40 GB	2 TB	\$7	\$14
Small	2 GB	2	60 GB	3 TB	\$12	\$22
Medium	4 GB	2	80 GB	4 TB	\$24	—
Large	8 GB	2	160 GB	5 TB	\$44	—
XLarge	16 GB	4	320 GB	6 TB	\$84	—
2XLarge	32 GB	8	640 GB	7 TB	\$164	—
4XLarge	64 GB	16	1,280 GB	8 TB	\$384	—

IPv6-only bundles cost about 30% less

A public IPv4 address is now the premium — the Linux nano tier drops from \$5 to \$3.50 without one.

Transfer overage is metered

Exceed the allowance and you pay per GB. Inbound traffic and traffic to other AWS services in-Region are free.

Three months free

Selected Linux/Unix IPv6 bundles at \$3.50, \$5 and \$10 carry a three-month introductory credit for new accounts.

What everything else costs

The rest of the catalogue is priced the same way — a flat monthly figure with an allowance attached.

Resource	Representative configuration	Price	What is included
Container service	Nano node — 0.25 vCPU, 512 MB	\$7 / month per node	500 GB transfer per service
Container service	XLarge node — 4 vCPU, 8 GB	\$160 / month per node	Scale = price × node count
Managed database	1 GB RAM, 40 GB SSD (standard)	\$15 / month	High-availability plans cost double
Object storage	5 GB storage, 25 GB transfer	\$1 / month	Free for the first 12 months
Object storage	250 GB storage, 500 GB transfer	\$5 / month	100 GB tier sits at \$3
Load balancer	Includes managed TLS certificate	\$18 / month	Certificate issue and renewal are free
CDN distribution	CloudFront edge caching	\$2.50 / month	50 GB/month free for 12 months
Block storage	Additional SSD disk	\$0.10 / GB / month	Attach up to 16 TB per instance
Snapshots	Instance or disk backup	\$0.05 / GB / month	Manual and automatic both charged
Static IP · DNS zone	IPv4 address · authoritative DNS	Free	IP billed only when left unattached

Read the pattern: AWS sells you an allowance, not a meter. That is the entire commercial difference between Lightsail and the rest of AWS.

Where the fixed price actually pays off

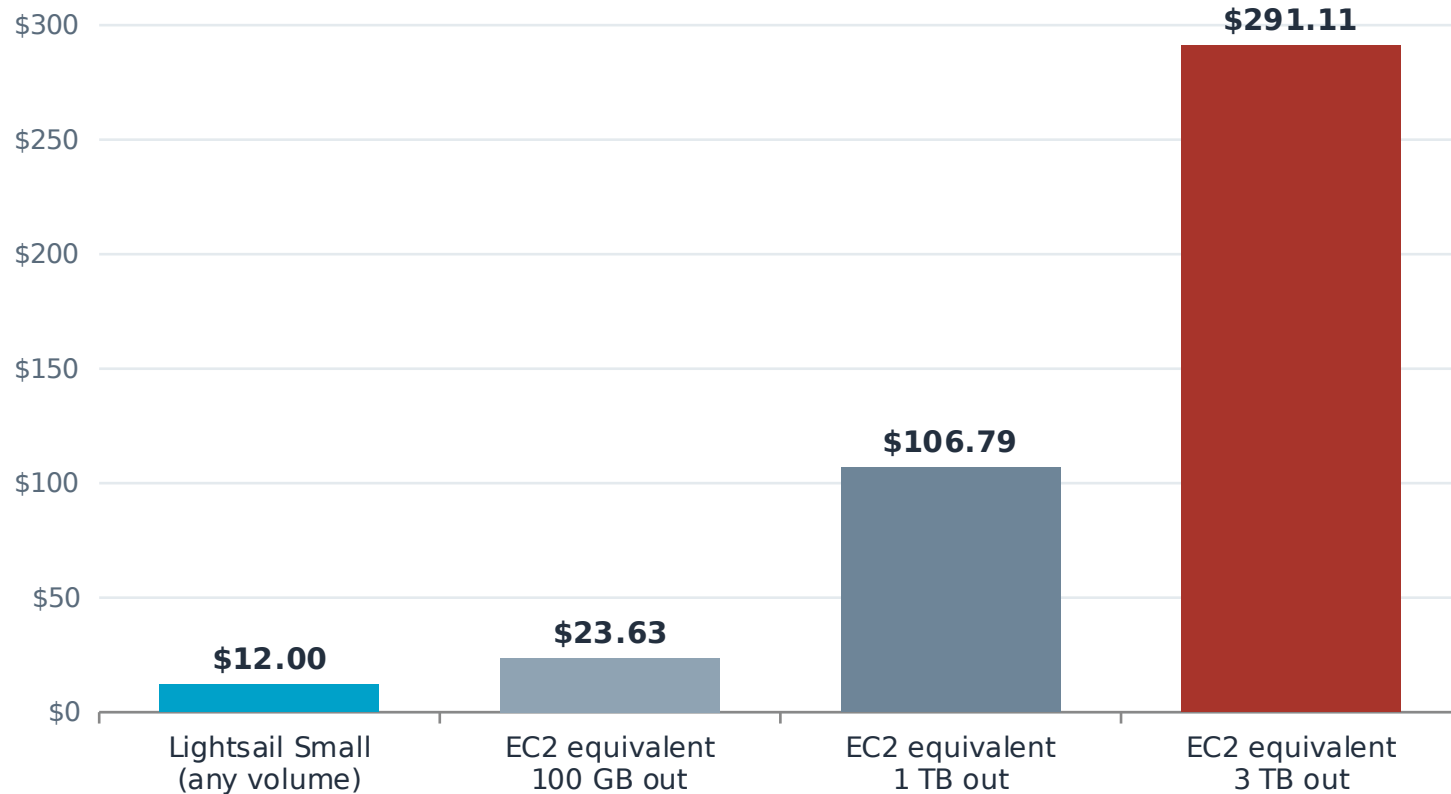


Figure 3 — Same workload, two billing models.

The assumption behind the bars

A Lightsail Small bundle is \$12/month for 2 GB RAM, 2 vCPU, 60 GB SSD and 3 TB of outbound transfer.

The comparison builds the same thing on EC2 at us-east-1 list prices: a t3.small on demand, a 60 GB gp3 volume, one Elastic IP, and egress at \$0.09/GB after the first 100 GB.

Read it honestly

Data transfer, not compute, drives the gap. At low traffic the two are close and EC2 wins back ground with Savings Plans. Lightsail's advantage is bandwidth included in the price — and a number you knew in advance.

Networking and security by default



Default-deny firewall

Every instance ships with its own firewall; only the ports the blueprint needs are open.



Free managed certificates

A load balancer issues and renews TLS certificates at no charge; instances can use Let's Encrypt.



Static IPv4, free in use

Attached addresses cost nothing. An unattached one is billed at \$0.005/hour to discourage hoarding.



Isolation and peering

Resources live in a Lightsail-managed VPC. One click peers it with your default VPC.

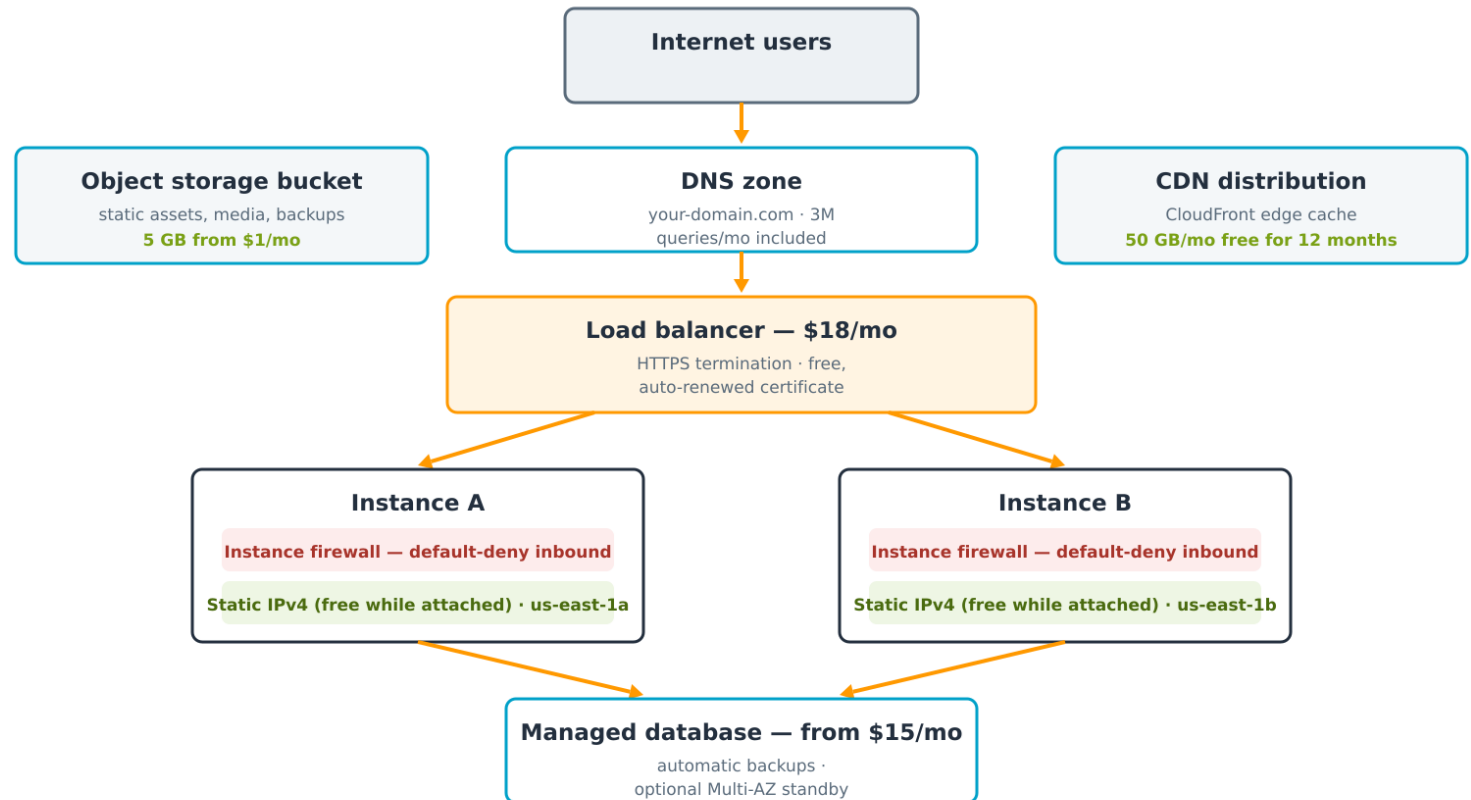


Figure 4 — A conventional highly available Lightsail deployment.

A simpler console does not shrink your security responsibility — patching the OS inside a blueprint is still your job.

Amazon Lightsail versus Amazon EC2

Dimension	Amazon Lightsail	Amazon EC2
Pricing model	Fixed monthly bundle, allowance included	Per-second metering across four separate meters
Choice of hardware	Eight general-purpose bundles plus two specialised families	More than 750 instance types
Maximum single instance	64 GB / 16 vCPU general purpose; 512 GB memory optimised	Up to 448 vCPU and 12,288 GB of memory
Networking	Lightsail-managed VPC; peering to the default VPC	Full VPC design: subnets, route tables, NAT, gateways
Automatic scaling	Not available for instances; containers scale by node count	EC2 Auto Scaling groups with policies and schedules
Discount instruments	None — the list price is the price	Reserved Instances, Savings Plans, Spot
Access control	Coarse; Lightsail-level permissions	Full IAM, resource policies, service control policies
Time to first server	Minutes, from a menu	Longer; several dependent resources must exist first
Best fit	Predictable small workloads, fast delivery, cost certainty	Elastic, large or compliance-heavy architectures

The two are not rivals: Lightsail is the on-ramp, EC2 the motorway, and the snapshot export makes the journey cheap.

COMPARISON

Lightsail against the wider market

Indicative entry pricing, August 2026. Figures are list prices and vary by region.

Platform	Entry price	You still manage	Distinguishing strength
Amazon Lightsail	\$5 / month (nano, dual-stack)	OS and application	Generous transfer allowance; a documented path onto full AWS
DigitalOcean Droplet	\$4 / month (512 MB); \$6 is the practical floor	OS and application	Clean developer experience and documentation
Linode / Akamai Nanode	\$5 / month (1 GB, 25 GB SSD, 1 TB)	OS and application	Strong price-to-memory ratio at the entry tier
Heroku dyno	\$5 Eco (sleeps) · \$7 Basic	Application only	True PaaS — no server to patch, but add-ons are billed separately
Azure App Service	Basic B1 around \$55 / month	Application only	Deep integration with the Microsoft estate

What actually separates them

On raw price per gigabyte of RAM, Lightsail is competitive rather than cheapest. It wins on two things a price list does not show: the included data-transfer allowance, which is where independent VPS bills usually break, and the fact that the resource underneath is ordinary AWS — so growing out of Lightsail is a migration inside one account rather than a change of vendor.

DEMONSTRATION

From nothing to a live HTTPS site



Live, HTTPS-secured WordPress site in under 10 minutes

The same result on raw EC2 needs VPC, subnet, route table, IGW, security group, EBS, Elastic IP and an install script.

Figure 5 — The complete deployment path for a production WordPress site.

Everything above is also a single API call:

```
aws lightsail create-instances --blueprint-id wordpress --bundle-id small_3_0
```

Where Lightsail earns its place



Websites and blogs

WordPress, Ghost and Drupal on a bundle whose bandwidth allowance covers ordinary traffic.



Development and test

Disposable environments that mirror production and are deleted when the sprint ends.



Small business applications

Line-of-business tools, booking systems and intranets for teams with no operations staff.



E-commerce storefronts

Magento and PrestaShop blueprints with a managed database and a CDN in front.



Learning and coursework

A genuine cloud server for the price of a coffee — the fastest route into AWS for a student.



Research computing

Lightsail for Research packages CPU and GPU workstations with the same fixed-price model.

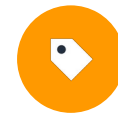
The common thread is a workload whose ceiling is known in advance. Lightsail is a poor fit the moment demand becomes genuinely unpredictable.

What you give up for the simplicity



Architectural limits

- No auto scaling for instances — capacity is a manual decision
- No Reserved Instances, Savings Plans or Spot pricing
- Coarse IAM; unsuitable where least-privilege access is audited
- Integration with the wider AWS estate goes through VPC peering only
- Entry bundles run on burstable CPU — sustained load throttles



Operational and commercial limits

- Default quotas: 20 instances, 5 static IPs and 6 DNS zones per account
- Exceeding the transfer allowance is billed per gigabyte
- Patching the OS and application inside a blueprint remains your job
- Bitnami-packaged blueprints are being retired through 2026-2027
- Not available in every AWS Region, unlike EC2

None of these is a defect. They are the deliberate cost of a fixed price — and each one is a signal that the workload may have outgrown the service.

Knowing when to leave — and how

STAY ON LIGHTSAIL

- Predictable, bounded traffic
- One to a handful of servers
- Team is small; ops time is scarce
- Cost certainty matters more than tuning

MOVE TO EC2

- Need auto scaling or spiky traffic
- Need > 64 GB RAM / 16 vCPU on one box
- Need IAM, custom VPC, Reserved Instances
- Need Savings Plans or Spot economics

The exit is built in

Take a snapshot → Export to Amazon EC2 → AWS creates an AMI + EBS snapshot → launch with the Upgrade to EC2 wizard
Your disk image is standard AWS. Migration is a copy, not a rewrite.

Figure 6 — The decision boundary, and the built-in migration route across it.

This is the case for Lightsail over an independent VPS: the exit is a supported AWS workflow, not a rebuild elsewhere.

CONCLUSION

Four things worth remembering

1

Lightsail sells predictability, not power

The bundle is a price you know before you deploy. Everything else follows from that one commercial decision.

2

It is PaaS-Lite, not PaaS

AWS pre-builds the operating system and runtime, then still gives you root. Convenience on day one, control on day two.

3

The saving is in the bandwidth

Compute is competitively priced; the included data-transfer allowance is where the fixed model genuinely pays.

4

Its limits are its exit signs

No auto scaling, no Savings Plans, coarse IAM. When those start to hurt, snapshot and export to EC2.

Thank you · Questions welcome

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All pricing and feature claims verified against these sources in August 2026.

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11. DigitalOcean, Akamai (Linode), Heroku and Microsoft Azure published price lists, consulted August 2026 for Slide 14.

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